

Open-NL: A Transparent Computational Testbed for Wide Dynamic Range Compression

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Abstract

Open-NL is an open-source computational testbed designed for the transparent modeling and evaluation of Wide Dynamic Range Compression (WDRC) prescriptive rules. Modern prescriptions like NAL-NL2 rely on extensive empirical regularizations to ensure clinical safety. These safeguards counteract the well-documented failures of pure intelligibility maximization. However, clinical software remains closed-source. This prevents researchers from isolating how specific heuristic safeguards interact.

Open-NL addresses this limitation. It couples a multi-start Nelder-Mead Speech Intelligibility Index (SII) optimizer with the Moore & Glasberg (2004) specific-loudness model. We benchmark this testbed across a $4^4 = 256$ -permutation heuristic sweep. To mitigate the numerical entrapment inherent to local simplex search on non-convex audiological surfaces, the framework deploys a robust 5-iteration multi-start initialization routine. Crucially, a sensitivity analysis reveals that the optimizer actively collides with these mathematical boundary conditions. Because the objective function is dominated by physiological loudness

constraints, modulating the underlying heuristic triggers yielded exceptionally tight response envelopes. This indicates that once the optimizer reaches the active penalty boundary, the underlying heuristic parameters are rendered largely irrelevant, effectively preventing runaway amplification but emphasizing the need to include binding penalty parameters in future sensitivity sweeps.

Standard monaural models systematically underestimate real-world binaural broadband summation. Therefore, this modeled loudness frontier is strictly illustrative rather than clinically definitive. When stripped of clinical heuristics, the optimizer dramatically inflates high-frequency gain in profound profiles. This behavior demonstrates the theoretical limits of pure mathematical optimization against real-world clinical bounds. Open-NL exposes these critical trade-offs within a fully inspectable framework. Ultimately, it provides the computational substrate for future calibration workflows, paving the way to replace static heuristic boundaries with individualized, distortion-aware objective functions.

I. INTRODUCTION

Manufacturer-agnostic prescriptions remain central to evidence-based hearing aid practice. While earlier investigations suggested that generic targets might outperform proprietary first-fit algorithms on patient preference and specific metrics (Valente et al., 2018), contemporary evidence indicates that aided speech recognition in noise often shows no significant difference across formulas (e.g., Cox et al., 2012). However, while formula choice has relatively modest intelligibility consequences in background noise, it drives substantial variations in overall loudness, making modeled loudness (quantified in sones per the Moore & Glasberg 2004

impaired loudness model) the primary dependent variable in prescriptive evaluation.

While the derivations of major algorithms like NAL-NL2 and DSL m[i/o] are published in detail, their software implementations remain closed-source. Audiological science has long recognized that unconstrained intelligibility maximization fails clinically without extensive empirical regularization. For example, the evolution from NAL-NL1 to NAL-NL2 required critical empirical corrections. These included global gain reductions and reduced compression ratios for severe losses. These safeguards were introduced specifically to counteract the aggressive over-amplification provoked by pure mathematical optimization (Keidser, Dillon, Carter, & O'Brien, 2012a).

Because clinical fitting software packages are compiled black boxes, researchers cannot isolate individual heuristic rules. It remains impossible to observe how specific safeguards interact within the optimization cascade. Existing open-source tools serve distinct, separate functional niches. The openMHA platform (Herzke et al., 2017) operates as a real-time signal processing master hearing aid rather than a target generator. The Cambridge CAM2/CAMEQ2-HF formulae (Moore et al., 2010) provide rigidly defined equation-based targets rather than a modular optimization sandbox. Finally, the Auditory Modeling Toolbox (AMT; Majdak et al., 2022) offers loudness modeling without native prescriptive inversion. Consequently, investigators cannot isolate a specific prescriptive heuristic within an optimization loop without reverse-engineering an entire proprietary engine.

Open-NL fills this gap. It provides a modifiable R substrate explicitly designed for the modular ablation of prescriptive heuristics. By coupling a Nelder-Mead desensitized SII optimizer to an integrated C++ specific-loudness engine, researchers can systematically disable, isolate, or invert individual heuristics. For instance, investigators can evaluate the

upward spread of masking when disabling the 30 dB conductive safety cap. This modular architecture aligns directly with evolving audiological frameworks, such as the multi-profile philosophy introduced in NAL-NL3 (Kitterick et al., 2026a).

Crucially, to benchmark this testbed without introducing confounding variables, the optimization layer is embedded within a strictly reproduced evaluation paradigm. The seven reference audiometric profiles, the Moore & Glasberg (2004) specific-loudness model, and the ANSI S3.5 SII metric utilized herein are a direct replication of the methodological framework established by Johnson and Dillon (2011). (Throughout this manuscript, “ANSI SII” refers to raw physical audibility, whereas “smoothed desensitized SII” or “complete desensitized SII” refers to audibility incorporating severe-loss desensitization and level distortion penalties). Because this physiological evaluation space is already established in the literature, the primary contribution of this manuscript is the transparent computational testbed itself. By exposing the behavior of numerical solvers within this standardized sandbox, we clarify a crucial distinction: while numerical solvers converge stably on any fixed objective space, theoretical WDRC target generation exhibits acute parameter sensitivity to uncalibrated heuristic boundaries, providing the computational infrastructure necessary to quantify and calibrate these interactions.

Clinical Safety and Usage Disclaimer

It is imperative to state unambiguously that Open-NL is strictly a computational research testbed and **must not be used for fitting hearing aids on human listeners in its current form**. Because the framework deliberately permits aggressive, over-prescriptive targets for boundary testing—such as utilizing an aggressive 60 dB HL severe-loss booster onset

that permits aggressively high-frequency targets for profound losses—it carries a significant risk of severe over-amplification. As established by Ching, Dillon, Katsch, and Byrne (2001), aggressive high-level targets in steeply sloping or profound losses are precisely where over-amplification risks are greatest, as the effectiveness of high-frequency audibility severely degrades as hearing loss worsens (desensitization). The hypotheses and targets generated by Open-NL represent extreme mathematical boundaries intended to trigger experimental loudness rejection in controlled research settings, not clinical solutions. Any future behavioral translation of this framework requires independent institutional review, with mandatory real-ear verification and strict, individualized loudness-tolerance limits implemented as absolute prerequisites.

II. ALGORITHM ARCHITECTURE

A. Prescriptive Rationale and Objective Function

The choice of objective function is the primary design decision in any prescriptive formula. It governs the fundamental trade-off between intelligibility and comfort. Historically, established rationales occupy distinct positions on this spectrum. NAL-NL2 maximizes speech intelligibility while constraining overall broadband loudness to be less than or equal to that of a normal-hearing listener (Keidser, Dillon, Carter, & O’Brien, 2012a). Conversely, DSL m[i/o] normalizes loudness across frequency to restore normal dynamic range perception (Scollie et al., 2005). Finally, CAMEQ/CAM2 aim to equalize loudness across frequency bands (Moore, Glasberg, & Stone, 2010).

Open-NL positions its prescriptive rationale as a *constrained intelligibility-maximizer*.

Its primary mathematical objective is the unconstrained maximization of desensitized SII. Rather than globally restricting this maximization to a static “normal-or-less” loudness boundary, Open-NL permits dynamic loudness growth. This growth continues until it strikes a U-shaped physiological ceiling (controlled via `cap_knots`; Section S.I.12). For severe losses, this penalty explicitly permits slightly higher-than-normal loudness in the mid-frequencies, where intelligibility yield is highest. However, it aggressively decelerates loudness growth at spectral extremes.

Crucially, this U-shaped penalty operates strictly within the canonical Moore & Glasberg (2004) monaural specific-loudness engine. It dynamically restricts modeled monaural sones. However, it does not—and mathematically cannot—account for the idiosyncratic binaural broadband loudness summation observed in hearing-impaired listeners. In normal-hearing auditory physiology, bilateral acoustic presentation produces a modest binaural loudness summation. This is typically modeled by a 2–6 dB level-dependent gain reduction.

However, robust psychoacoustic evidence demonstrates a stark contrast in impaired ears. Binaural broadband summation in hearing-impaired populations averages ~13 dB higher than in normal-hearing listeners. This represents an unmodeled factor of $\approx 2.4\times$ in linear sones (Denk et al., 2025; Moore et al., 2014; Oetting et al., 2016, 2017). About 40% of hearing-impaired listeners (in a sample of 180) exhibit excess summation far exceeding the normal range. Individual summation values span a massive -10 to +40 dB envelope.

Standard monaural and narrowband loudness models cannot predict this broadband suprathreshold phenomenon from the pure-tone audiogram alone. Therefore, an algorithm optimized strictly beneath a monaural ceiling becomes structurally anti-conservative when translated to bilateral fittings. Consequently, Open-NL’s U-shaped loudness constraint must

be interpreted strictly as an illustrative computational boundary for single-ear simulation,
rather than an empirical safety guarantee for bilateral clinical use.

B. Methods and Development

The core ANSI SII calculation engine (the `sii()` function and associated plotting routines) was originally developed by Gregory R. Warnes for earlier package versions. Maintenance transferred to the current author with version 1.1.0, at which point all subsequent Open-NL prescriptive logic, clinical heuristics, and WDRC mathematical implementations—including `open_nl()` and `calculate_loudness()`—were developed by the author as original contributions.

Declaration of Generative AI and AI-assisted technologies in the research process: In accordance with AIP Publishing guidelines, the author explicitly discloses the use of Gemini 3.1 Pro (DeepMind, Google LLC) during the preparation of this work as an interactive programming and copyediting assistant. The AI was utilized strictly to refactor C++ and R algorithms, generate data visualizations, and condense manuscript prose to adhere to JASA formatting standards. After using this tool, the author rigorously reviewed and edited all outputs, taking full accountability for the underlying algorithm design, theoretical hypotheses, and final manuscript content. Specifically, to guarantee computational integrity, all AI-assisted algorithmic refactoring was systematically verified by the author via exact numerical regression testing against pre-refactor outputs across the seven canonical audiometric profiles, confirming absolute mathematical parity during translation.

C. Algorithmic Pipeline and Execution Cascade

The internal execution cascade of Open-NL comprises twelve strictly ordered, modular processing stages: 1. **Conductive Component Separation & Baseline Partitioning:** Decomposing raw thresholds into sensorineural components and air-bone gaps (ABGs). 2. **Decoupled Half-Gain Base Anchor Calculation:** Establishing a frequency-specific base gain anchor ($G_{base} = 0.46 \cdot \text{HTL}_{sn} + C_{interp}$) decoupled from broadband PTA. 3. **Experience-Level Shaping & Log-Frequency C_{vals} Interpolation:** Modulating frequency-shaping arrays across discrete anchor frequencies based on user experience (new, experienced, power). 4. **Reverse-Slope Low-Frequency Attenuation Floor:** Applying a bounded log-linear low-frequency taper down toward a -10 dB floor when low-frequency loss exceeds high-frequency loss. 5. **Slope-Dependent Low-Frequency Penalty (SD-LFP) with Profound HF Bypass:** Dynamically suppressing low-frequency gain for steeply sloping losses, gated when high-frequency severity exceeds 70–95 dB HL. 6. **Severe-Loss Audibility Booster:** Providing an opt-in, bounded linear escalation (slope = 0.15) for severe thresholds. 7. **Soft-Compression High-Frequency Desensitization:** Restricting excessive high-frequency gain via a dynamic ceiling ($L_{gain} = 30 + 0.4 \cdot \max(0, \text{HTL}_{sn} - 60)$) and 2:1 soft compression. 8. **Dynamic Range Mapping (LDL Squeeze):** Attenuating gain by 0.2 dB/dB of reduced dynamic range and shifting baseline compression ratios. 9. **Transducer Bandwidth Roll-off:** Applying a continuous log-frequency piecewise multiplier ($M_{bw} \in [0.5, 1.0]$) to suppress unstable edge frequencies (≤ 250 Hz and ≥ 6000 Hz). 10. **Multi-Channel WDRC Mapping, LTASS Pivot, and Demographic Adjustments:** Calculating dynamic compression ratios, variable compression thresholds, piecewise linear-compressive splines, and

demographic offsets. 11. **Acoustic Venting, Coupling Loss, and Receiver Saturation**

Limits: Integrating real-ear insertion loss vectors across 10 coupling types with a feedback

safety floor, and capping MPO/SSPL90 receiver limits at 120 dB SPL. 12. **Embedded**

Nelder-Mead Simplex Optimization & Physiological Loudness Ceilings: Iteratively

refining multi-level gain curves against a multi-term objective loss function incorporating U-

shaped specific-loudness caps, compression ratio ceilings, and spectral smoothness constraints.

The exact mathematical formulations, closed-form piecewise equations, and complete

parameter specifications for all twelve stages are provided in full detail in the accompanying

Supplementary Material.

D. Consolidated Constants and Evidentiary Asymmetry

Because Open-NL functions as a modifiable computational testbed, several structural

parameters remain mathematically uncalibrated: the 0.46 gain anchor, 0.15 severe-loss

booster slope, 70 dB HL booster onset (or 60 dB HL in aggressive mode), 15 dB slope

trigger, 20 dB taper width, -10 dB reverse-slope floor, 70 dB HL bypass, 30 dB/0.4 L_{gain}

constraint, 0.2 dB/dB dynamic range squeeze, 75% air-bone-gap restoration fraction, and

1.5:1 Comfort-in-Noise (CIN) compression clamp (Table S1).

Crucially, an inspection of these heuristics reveals a profound asymmetry in evidentiary

support. Several prominent constants—most notably the 75% air-bone-gap restoration rule

and the 1.5:1 Comfort-in-Noise clamp—are pragmatic engineering choices without direct

empirical derivation. The 75% ABG fraction, while standard in clinical prescriptive software

(Johnson, 2013a) to avoid receiver saturation and MPO clipping, has never been empirically

established against patient preference or speech recognition. Similarly, the 1.5:1 CIN clamp

is an asserted heuristic inspired by NAL-NL3 (Kitterick et al., 2026b) to mitigate listening fatigue, but lacks independent perceptual validation. In sharp contrast, the 3.0:1 Compression Ratio (CR) upper ceiling enforced across the WDRC stages and optimizer loss function represents the best-supported constant in the framework. This boundary is firmly anchored in the extensive empirical literature by Pamela Souza and colleagues (Souza, 2002; Souza, Jenstad, & Boike, 2006), which demonstrates that compression ratios exceeding $\sim 3.0:1$ cause severe temporal envelope flattening, loss of acoustic contrast, and speech-in-noise deficits. Documenting this asymmetry prevents conflating validated psychoacoustic limits with arbitrary engineering heuristics.

E. Framework for Principled Calibration

For a computational testbed to provide lasting inferential value, demonstrating parameter instability must be paired with a rigorous calibration pathway. Future research can utilize Open-NL within a global stochastic optimization framework (e.g., genetic algorithms or particle swarm optimization) to systematically calibrate these heuristic gates. In this architecture, an outer optimization loop searches the multidimensional space of heuristic constants (e.g., $G_{base} \in [0.3, 0.6]$, booster onset $\in [50, 80]$ dB HL, low-frequency slope $\in [0, 0.5]$). For each candidate vector, the inner Open-NL engine generates discrete WDRC targets across a representative clinical corpus (e.g., NHANES). These targets are evaluated through a time-domain acoustic simulation pipeline (e.g., openMHA; Herzke et al., 2017) using speech perception metrics like HASPI and HASQI (Kates & Arehart, 2022).

Crucially, the outer-loop objective function must incorporate an asymmetric, veto-based cost function: any parameter configuration violating individualized broadband loudness

tolerance (trueLOUDNESS; Oetting et al., 2017) in simulated outlier listeners must incur an overwhelming penalty. As detailed in Section II.A, because excess binaural broadband summation cannot be predicted from pure-tone thresholds, static 2–6 dB bilateral corrections are structurally inadequate. Subordinating population-level intelligibility maximization to individualized physiological safety limits transforms prescriptive derivation into a reproducible, distortion-aware computational science.

TABLE I. Comprehensive Enumeration of Open-NL Free Parameters and Evidentiary Derivation.

	Specific Free	Default / Evaluated	Evidentiary Support
Parameter Category	Parameters	Value	& Derivation
Objective	Loudness Cap Knots	L_{cap} vectors (Section	Derived from
Penalties	(<code>cap_knots</code>)	S.I.12)	normal/impaired
			physiological
			loudness growth
			models.
	Optimizer Penalty	$\lambda_1 = 450, \dots, \lambda_8 =$	Pragmatic
	Weights (λ_{1-8})	20 (Sec S.I.12)	engineering
			constraints balancing
			target convergence.

	Specific Free	Default / Evaluated	Evidentiary Support
Parameter Category	Parameters	Value	& Derivation
Prescriptive Anchors	CR Soft Penalty	$\leq 3.0 : 1$	Strong empirical
	Target (P_{cr})	Compression Ratio	derivation: Souza (2002) speech degradation limits.
	Base Gain Anchor	0.46	Uncalibrated
	(G_{base})		midpoint balancing half-gain rules and preference data.
	New-User Offset	0 to -6 dB based on	Assumed heuristic
	(Δ_{exp})	PTA	approximating acclimatization preferences.
	Severe-Loss Booster	0.15 slope / 70 dB	Assumed heuristic
	(Slope / Onset)	HL onset	assisting profound loss without explosive recruitment.

	Specific Free	Default / Evaluated	Evidentiary Support
Parameter Category	Parameters	Value	& Derivation
Compression Limits	ABG Restoration Fraction	75% (Linear)	Engineering choice preventing hardware saturation (Johnson, 2013a).
	Compression	30 to 45 dB SPL	Pragmatic limit
	Kneepoint (CT) Range		aligning with standard real-world WDRC kneepoints.
	High-Frequency	Base 30 dB, Slope	Heuristic managing
	Soft-Compression	0.4	upward spread of masking in severe losses.
	DR Squeeze / CR	0.2 dB/dB, 0.02	Heuristics fitting
	Shift	CR/dB	speech envelope into reduced physiological space.
	MPO/LDL Predictor	See Stage 8	Statistical
	Coefficients		predictions based on population normative UCL datasets.

	Specific Free	Default / Evaluated	Evidentiary Support
Parameter Category	Parameters	Value	& Derivation
Frequency Limits	Reverse-Slope Floor	-10 dB	Cautious heuristic preventing masking of intact basal cochlear units.
	Dead Region /	30 dB/oct past $1.7f_e$,	Standard transducer
	Transducer Roll-off	w_{bw}	limits and dead-region literature (Moore, 2001).
	Acoustic Coupling Loss	Occluded/Vented Vectors (Table S4)	Deterministic physical hardware measurements.

III. ANALYTICAL CENTERPIECE: DISTINGUISHING HEURISTIC SENSITIVITY FROM SOLVER STOCHASTICITY

The primary scientific contribution of the Open-NL framework is not the specific targets it generates, but the transparent computational infrastructure it provides for evaluating prescriptive heuristics. Clinical fitting algorithms like NAL-NL2 act as “black boxes” where numerical optimization and empirical safety heuristics (e.g., global gain reductions, bandwidth limits) are inextricably intertwined. Open-NL fills this methodological gap by providing an ablatable, inspectable testbed that mathematically isolates the behavior of unregularized

optimization on the physiological loudness landscape.

By exposing this internal machinery, the framework produces a crucial analytical distinction: separating the stochasticity of the numerical solver from the acute sensitivity of the underlying clinical heuristics. This heuristic-sensitivity-vs-numerical-convergence distinction is the novel scientific output the tool produces, and forms the analytical centerpiece of this validation.

A. Heuristic Parameter Sensitivity vs. Numerical Convergence Stability

1. Multi-Parameter Sensitivity Sweep To isolate heuristic sensitivity from numerical solver stochasticity, an ANOVA variance decomposition (reporting η^2 effect sizes) was executed across 256 permutations of four primary algorithmic constraints: the base gain anchor (0.40 to 0.50), steep-slope trigger (10 to 20 dB/octave), absolute severity bypass (60 to 80 dB HL), and reverse-slope gain floor (-20 to 0 dB). These 256 unique parameter permutations were evaluated independently on each of the most topologically unstable profiles (A2, A4, A5)—yielding 768 total permutation runs (256 per profile)—executed with the aggressive 60 dB HL booster mode engaged to directly probe boundary interactions.

TABLE II. ANOVA Variance Decomposition of Heuristic Parameters (65 dB SPL Input). *Note: Generated using the aggressive 60 dB HL booster onset. The ANOVA decomposition explicitly separates variance explained by clinical heuristics from residual solver stochasticity.*

	Median Desensitized	Median Loudness	ANOVA Dominant
Profile	SII [Min, Max]	[Min, Max]	Factor (η^2)
A2	0.87 [0.82, 0.88]	4.43 [3.38, 4.44]	Anchor (61.1%)
A4	0.64 [0.62, 0.66]	5.27 [5.23, 5.28]	None (Total Variance < 0.1 sones)
A5	0.45 [0.45, 0.51]	4.28 [4.26, 4.37]	None (Total Variance < 0.2 sones)

246 The resulting variance (**Table II, Figure 1**) illustrates mechanistically how strict mathe-
 247 matical boundary conditions dominate the objective landscape. The tight response envelope is
 248 not evidence of a flat optimization space, but rather demonstrates that the optimizer actively
 249 collides with the binding physiological loudness constraint. For example, the interpolated
 250 loudness caps for profiles A1, A3, and A5 effectively dictate the final modeled loudness to
 251 within 0.1 sones. Because the objective function is entirely bound by these hard distortion
 252 penalties, modulating the underlying heuristic triggers failed to produce any significant main
 253 effect (e.g., A5 fluctuating narrowly between 4.26 and 4.37 sones purely due to higher-order
 254 interactions). This indicates that once the active penalty wall is reached, the underlying
 255 heuristic parameters are rendered practically irrelevant. While this successfully prevents the
 256 massive runaway amplification typical of historically unregularized intelligibility optimization,
 257 any future claims of comprehensive parameter robustness must be tested against a sweep
 258 that formally includes these binding penalty variables.

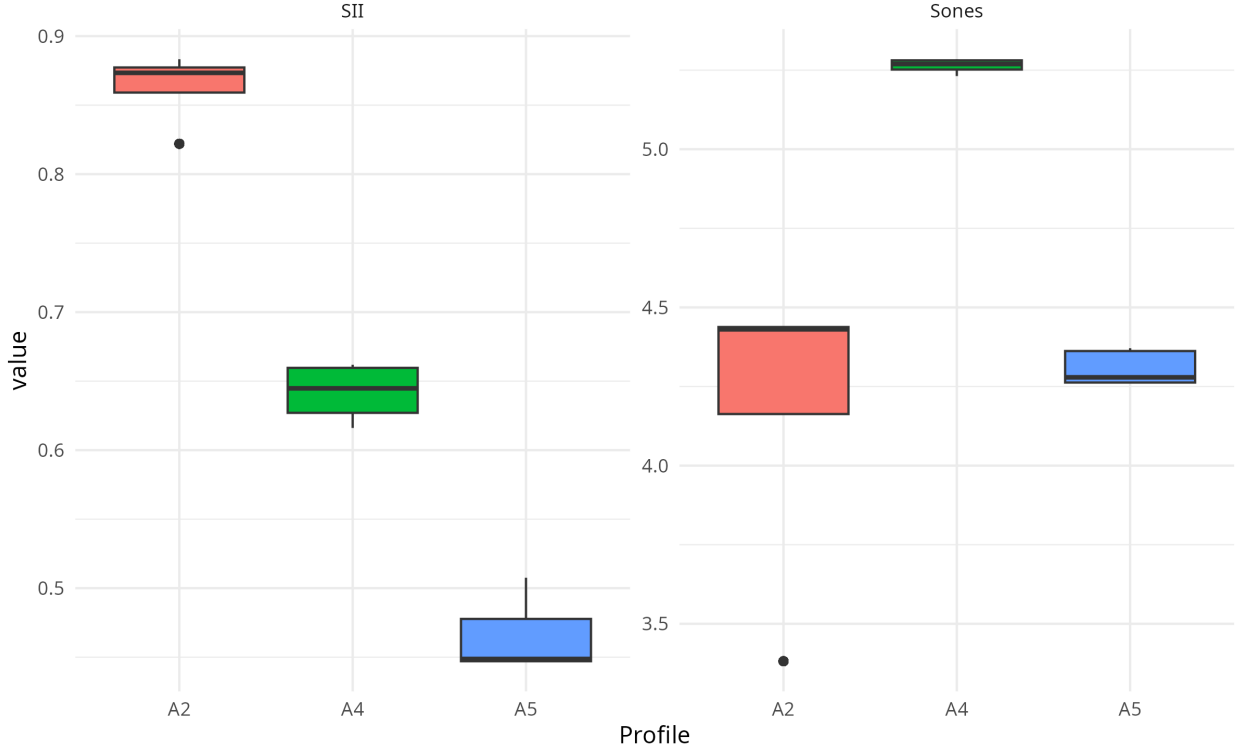


Figure 1. Distribution of resulting ANSI SII scores and physiological loudness penalties across 768 permutations (256×3 profiles) for clinical profiles A2 (Reverse Slope), A4 (Severe), and A5 (Profound), illustrating how the active physiological loudness constraint dominates the optimization space, severely restricting the variance caused by heuristic parameter modifications.

2. Numerical Convergence Stability: Nelder-Mead Limitations and Future

Stochastic Solvers A sharp distinction must be maintained between **heuristic parameter sensitivity** (target shifts resulting from altered clinical rules; Section III.A.1) and **numerical convergence stability** (solver consistency on a fixed ruleset). In non-linear optimization, the topology of hearing aid fitting targets is notoriously ill-behaved. The objective landscape contains narrow, curved valleys, non-differentiable step boundaries (e.g.,

severe-loss booster onsets and air-bone gap restorations), and sharp penalty cliffs imposed by dynamic physiological loudness ceilings. On such non-convex, multimodal surfaces, local downhill solvers like the Nelder-Mead simplex algorithm are notoriously prone to premature stagnation, simplex collapse, and entrapment in shallow local extrema.

Indeed, unconstrained Nelder-Mead search from disparate flat initializations (e.g., -10 dB vs. +10 dB) can deviate by up to 0.5 sones or 0.05 SII. To mitigate this, Open-NL deploys a 5-iteration multi-start routine—seeding the initial simplex with the NAL-R target and executing four additional randomized restarts.

Because the ANOVA decomposition demonstrated that the heuristic rules had minimal main effects on the steeply sloping profiles (A4, A5), the absolute variance observed across the sweep was exceptionally tight (a total range of just 0.05 sones for A4). However, this residual variance represents a convolution of highest-order heuristic interactions and solver stochasticity. Consequently, this sweep cannot strictly isolate pure numerical stability. Future investigations must formally evaluate Nelder-Mead solver consistency by executing dedicated Monte Carlo initializations with fixed parameter vectors across hundreds of varying RNG seeds.

Furthermore, while the algorithm tightly constrains profiles like A4 and A5 against the physiological loudness cap, other profiles exhibit significant heuristic sensitivity. For example, profile A2 experiences wide target swings (3.38 to 4.44 sones) systematically driven by the anchor heuristic (61.1% variance). Because physiological optimization constraints often create flat topological plateaus near the optimal basin, divergent parameter configurations can theoretically produce functionally equivalent objective scores. Future implementations analyzing parameter identifiability should transition to modern global stochastic optimization

algorithms:

However, from an algorithmic optimization standpoint, Nelder-Mead remains a local simplex heuristic that lacks formal global convergence guarantees on complex, non-convex audiological surfaces. While multi-start seeding with NAL-R provides a practical, computationally efficient substrate for local ablation testing within R, **future iterations of prescriptive target optimizers should transition to modern global stochastic optimization algorithms:**

- 1. Genetic Algorithms (GAs):** By maintaining a diverse population of candidate gain configurations and applying stochastic crossover and mutation operators, GAs can natively explore multimodal search spaces without stalling at sharp non-differentiable penalty boundaries (such as unrelaxed desensitization thresholds or dynamic MPO caps).
- 2. Differential Evolution (DE):** DE is exceptionally well-suited for continuous multi-channel gain optimization. Its vector-difference mutation mechanism enables self-adaptive search across non-convex landscapes, effectively escaping the local attractor basins that trap simplex algorithms in steeply sloping audiograms.
- 3. Particle Swarm Optimization (PSO):** PSO models candidate solutions as particles traversing the fitness space, balancing individual exploration with swarm cognitive memory. This provides rapid convergence across multi-channel parameter sweeps while maintaining robust resistance to local minima.

Crucially, adopting population-based global stochastic solvers eliminates the need for artificial mathematical relaxations (such as the smoothed $K'_{smoothed}$ desensitization approximation), allowing optimizers to evaluate raw, discontinuous physiological boundaries directly. Furthermore, global solvers scale naturally to high-dimensional multi-channel optimization, providing the robust algorithmic substrate necessary for outer-loop calibration frameworks

(Section II.E). Nevertheless, as noted by Dao et al. (2021), physical real-ear-to-coupler differences (RECD) and acoustic coupling leakages cause real-world fittings to deviate most severely from theoretical targets at thresholds above 85 dB HL, meaning numerical convergence on mathematical audiograms must not be misconstrued as physical clinical reliability.

B. Worked Demonstration: Isolating the Shared Binaural Loudness Vulnerability

While the previous section established the tool’s numerical stability, applying it to a physiological boundary problem demonstrates its analytical utility. A central problem in modern audiology is that standard clinical models operate on a monaural basis, failing to account for idiosyncratic binaural broadband loudness summation. The fact that established prescriptions like NAL-NL2 share this vulnerability to unpredictable binaural loudness is precisely why having an open, parameterizable model like Open-NL is valuable: it provides a testbed to isolate and simulate the effect of this field-wide blind spot without it being buried under opaque empirical corrections.

The following comparison between Open-NL and NAL-NL2 is therefore not presented as a finding about Open-NL’s amplification superiority, but as a worked demonstration of the framework’s ability to expose structural vulnerabilities that the field cannot solve from the pure-tone audiogram alone.

To benchmark Open-NL without confounding variables, the optimization layer is evaluated within the established 7-profile paradigm of Johnson and Dillon (2011), comparing targets directly against NAL-NL2 version 2 software (National Acoustic Laboratories, Sydney, Australia) for soft (50 dB SPL), conversational (65 dB SPL), and loud (80 dB SPL) speech inputs. All NAL-NL2 targets were extracted using an 18-channel compression architecture with

adaptive time constants, occluded BTE #13 tubing, and supra-aural headphone thresholds. Crucially, to isolate the pure mathematical objective function without demographic artifacts, both Open-NL and NAL-NL2 targets were generated using identical baseline configurations: **Bilateral, Adult, Unknown Gender, Experienced user, and Non-tonal language.** Both sets are reported strictly as Real-Ear Insertion Gain (REIG) in the identical acoustic reference plane with matched occluded coupling. To ensure absolute scoring parity, both Open-NL and NAL-NL2 final targets were mathematically evaluated through the exact same objective metric engine: the ANSI/ASA S3.5-1997 (R2024) standard, utilizing the critical-band calculation procedure (21 bands) and the Normal vocal effort Long-Term Average Speech Spectrum (LTASS). Restricting comparisons to NAL-NL2 provides a standardized, universally recognized clinical baseline, avoiding the artifacts of surrogate target estimators for alternative proprietary formulae.

Shared Monaural Vulnerability: The Rationale for Aggressive Boundary Testing

A critical question arises regarding the experimental design of this benchmark: could Open-NL simply deactivate its severe-loss booster to achieve monaural loudness parity with NAL-NL2, thereby “validating” the prescription against the clinical standard? Indeed, in default conservative mode (booster onset at 70 dB HL or disabled), Open-NL yields monaural loudness values closely aligned with NAL-NL2 across primary sensorineural profiles. However, using conservative parity to claim clinical validation would obscure the central theoretical lesson of this computational testbed.

Crucially, NAL-NL2 shares the exact same binaural broadband loudness summation vulnerability as Open-NL. NAL-NL2 applies the identical, standard level-dependent 2–6

361 dB bilateral reduction derived from normal-hearing listeners, failing equally to account for
362 the excess broadband summation documented in hearing-impaired populations (see Section
363 II.A). The reason NAL-NL2 does not trigger widespread clinical loudness rejection is not
364 because its underlying loudness model is physiologically complete, but because its empirical
365 derivations incorporated heavy, post-hoc regularizations—including global gain reductions (-2
366 dB for females, -3 dB for new users), compressed dynamic range ceilings, and conservative
367 high-frequency roll-offs—combined with routine clinical reliance on the ± 15 dB volume
368 control in fitting software (Keidser et al., 2012a).

369 By intentionally executing this evaluation with the aggressive 60 dB HL booster mode
370 engaged, Open-NL deliberately strips away these empirical dampers. This stress-testing
371 reveals that when an unregularized numerical optimizer operates strictly beneath standard
372 monaural loudness models, it aggressively pushes high-frequency gain and exploits the objec-
373 tive function, producing targets that are structurally anti-conservative for bilateral fittings.
374 Rather than merely mimicking NAL-NL2’s output envelope, Open-NL’s diagnostic value lies
375 in demonstrating that relying exclusively on static monaural loudness assumptions introduces
376 substantial anti-conservative risks for the significant cohort of patients (30–40%) exhibiting
377 excess bilateral summation. This strongly suggests that future prescriptive frameworks
378 (e.g., NAL-NL3 and Open-NL v2) cannot resolve bilateral loudness through static audio-
379 metric equations, but must directly integrate individualized broadband loudness scaling
380 (*trueLOUDNESS*; Oetting et al., 2017) into their objective functions.

381 Two critical methodological boundaries govern this evaluation: 1. **The Constrained**
382 **Metric Tautology Warning:** A fundamental circularity exists when evaluating an optimiza-
383 tion algorithm against the identical metric it was tuned to maximize. Because Open-NL’s

objective function seeks to maximize desensitized SII, reporting higher SII values relative to regularized formulae (like NAL-NL2) is generally expected. However, examining the full distribution in Table III reveals that Open-NL’s objective exploitation manifests through three distinct mechanical pathways: - **Active Constraint Inversions (A4, A5)**: As previously noted, Open-NL yields *lower* desensitized SII than NAL-NL2 for A4 (0.62 vs. 0.68) and A5 (0.44 vs. 0.54). Here, the U-shaped physiological loudness cap forces the solver to explicitly sacrifice theoretical audibility to prevent catastrophic loudness growth. - **Solver Entrapment and Pareto Domination (A1, A2)**: For profile A2, Open-NL is strictly Pareto-dominated by NAL-NL2: it yields a lower desensitized SII (0.77 vs. 0.78) while being drastically louder (4.44 vs. 3.43 sones). This occurs because Nelder-Mead becomes trapped against the active 4.44-sones penalty wall. Lacking the ability to accept temporary uphill loss to traverse the non-convex landscape, the local simplex solver fails to locate NAL-NL2’s objectively superior gain allocation. This starkly demonstrates the necessity of transitioning to global stochastic solvers (Section III.A.2). - **Inefficient Objective Exploitation (A6, A7)**: For profile A6, Open-NL blindly trades a massive 1.36 sones of excess loudness to buy a marginal +0.04 increase in SII. For A7 (a purely deterministic conductive correction), Open-NL yields +0.62 sones for zero additional SII gain. This behavior highlights the inherent danger of pure unregularized optimization: algorithms will indiscriminately sacrifice patient comfort for statistically insignificant fractions of objective audibility unless heavily penalized.

Ultimately, stationary band-importance metrics like ANSI S3.5 and desensitized SII are blind to dynamic temporal envelope distortion, channel cross-talk, and phase distortion induced by aggressive compression ratios. In auditory science, genuine, independent, distortion-aware speech perception evaluation requires waveform-level biophysical models: -

HASPI (Hearing Aid Speech Perception Index; Kates & Arehart, 2022): Accurately simulates peripheral auditory processing, basilar membrane compression loss, auditory nerve firing rates, and envelope modulation integrity. - HASQI (Hearing Aid Speech Quality Index; Kates & Arehart, 2022): Evaluates non-linear harmonic distortion, envelope fidelity, and spectral fine-structure cross-correlation between aided and reference speech signals. Computing HASPI and HASQI requires convolving continuous speech (.wav) through a time-domain dynamic range compression engine (such as openMHA; Herzke et al., 2017). Because Open-NL currently operates strictly at the steady-state prescriptive target level (Johnson & Dillon, 2011), it lacks the native time-domain waveform processing required to compute HASPI and HASQI. Consequently, the objective metric differentials reported in Table III and Figure 3 are presented as theoretical bounds tests—quantifying the mathematical consequences of removing clinical heuristics—rather than as direct clinical superiority claims.

2. Binaural Loudness Summation and the Collapse of Monaural Frontiers: The comparative loudness evaluations are fundamentally bounded by the limitations of monaural auditory modeling. While standard clinical software applies a nominal 2 to 6 dB bilateral gain reduction, this static correction reflects normal-hearing physiology and fails catastrophically for broadband speech in impaired listeners. As detailed in Section II.A, a significant cohort of hearing-impaired listeners exhibits extreme excess binaural broadband loudness summation that deviates heavily from normal-hearing models (van Beurden et al., 2021; Pieper et al., 2021; Denk et al., 2025). Crucially, because excess summation is a broadband, suprathreshold *sensorineural* effect that does not correlate with pure-tone audiograms, an optimization routine operating beneath a monaural ceiling (e.g., 4.32 sones for profile A5) appears mathematically safe in isolation, yet predictably collapses into acute acoustic intolerance when fitted

bilaterally. Furthermore, there is no physiological basis for applying such excess summation models to purely conductive etiologies (e.g., A7). Because applying a fixed scalar to monaural outputs assumes normal-hearing loudness growth—the exact structural flaw this framework critiques—Table III reports canonical single-ear monaural loudness exclusively. True bilateral predictions require propagating the dynamic full-range signal through a non-linear binaural loudness engine.

To execute this evaluation natively in R, the **SII** package implements a fast C++ port of the canonical Moore & Glasberg (2004) stationary specific-loudness model via **Rcpp**. (Note: As stated in the AI Declarations, the internal R-to-C++ translation was verified to absolute mathematical parity via numerical regression testing).

To externally cross-check this stationary implementation, native C++ predictions were evaluated against the Auditory Modeling Toolbox (AMT; Majdak et al., 2022) across 45 discrete test points (5 profiles $\times$ 9 input levels from 50 to 90 dB SPL). Specifically, Open-NL’s stationary outputs were compared against AMT’s **bramslow2004** function. It is critical to note that this is not a direct port comparison, but rather a cross-algorithmic validation: Open-NL integrates the steady-state algebraic power spectrum directly, whereas AMT’s **bramslow2004** runs a physical 2400-sine-wave stimulus with random phases through a simulated time-domain digital filterbank. Because the time-domain model inherently captures the transient crest-factor peaks of the crest-factored noise waveform, exact machine-precision agreement is mathematically impossible.

Despite these fundamentally distinct modeling pathways (stationary spectral integration vs. dynamic time-domain waveform simulation), the models exhibited excellent approximate convergence: a mean bias of just +0.23 sones and a Mean Absolute Error (MAE) of 0.39

453 sones (**Figure 2**).

Bland-Altman Agreement: Open-NL C++ vs AMT glasberg2002

Difference (C++ - AMT)

Mean Loudness (Sones)

454

455 *Figure 2. Difference plot demonstrating excellent approximate convergence between Open-NL's*
456 *stationary spectral C++ engine and AMT's dynamic time-domain simulation (bramslow2004)*
457 *across 45 canonical evaluation points.*

458 For mixed and conductive profiles (A6, A7), direct AMT benchmarking was omitted
459 because canonical AMT lacks native air-bone gap parameters, whereas our C++ engine
460 algorithmically extends the model to treat the conductive component as a linear pre-cochlear
461 attenuator, in accordance with standard audiological principles (Dillon, 2012).

462 **TABLE III. Diagnostic Demonstration of Objective Exploitation: Monaural**
463 **Loudness (Sones) and Desensitized SII across A1-A7 Audiograms (65 dB SPL**
464 **Input).** *Note: Open-NL targets are presented for both Conservative (booster onset 70 dB HL)*

465 and Aggressive (onset 60 dB HL) modes for steeply sloping profiles (A4, A5). For profiles not
 466 exceeding 60 dB HL (A1-A3, A6-A7), the booster does not engage and outputs are identical
 467 across modes, designated simply as ‘Open-NL’. The comparative columns illustrate theoretical
 468 boundaries of optimization against clinical anchors.

				Monaural
				Loudness
Profile	Formula	ANSI SII	Desensitized SII	(Sones)
A1	NAL-NL2	0.82	0.76	4.29
A1	Open-NL	0.82	0.76	4.44
A2	NAL-NL2	0.84	0.78	3.43
A2	Open-NL	0.85	0.77	4.44
A3	NAL-NL2	0.71	0.67	3.92
A3	Open-NL	0.72	0.67	4.20
A4	NAL-NL2	0.71	0.68	6.09
A4	Open-NL	0.58	0.55	5.22
	(Conservative)			
A4	Open-NL	0.65	0.62	5.27
	(Aggressive)			
A5	NAL-NL2	0.57	0.54	5.53
A5	Open-NL	0.47	0.44	4.26
	(Conservative)			

Profile	Formula	ANSI SII	Desensitized SII	Monaural
				Loudness (Sones)
A5	Open-NL (Aggressive)	0.47	0.44	4.32
A6	NAL-NL2	0.79	0.75	2.12
A6	Open-NL	0.85	0.79	3.48
A7	NAL-NL2	0.97	0.92	1.15
A7	Open-NL	0.97	0.92	1.77

469 **TABLE IV. Insertion Gain Targets (dB) across A1-A7 Audiograms (65 dB**
 470 **SPL Input).** *Note: Open-NL targets are presented for both Conservative and Aggressive*
 471 *modes for A4 and A5. Targets illustrate how unconstrained desensitized SII maximization*
 472 *allocates high-frequency gain relative to regularized formulae. Profile A7 is fully deterministic*
 473 *(0.75 x 50 dB = 37.5 dB) and is included strictly as an arithmetic sanity check.*

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A1	NAL-	0.0	0.0	7.3	12.1	18.0	19.1
	NL2						
	Open-NL	0.0	7.2	15.8	18.4	22.0	12.8
A2	NAL-	10.1	9.3	12.2	8.3	3.9	4.0
	NL2						
	Open-NL	11.8	18.0	20.4	13.8	8.2	2.5

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A3	NAL-	0.0	0.0	9.9	16.8	20.7	21.6
	NL2						
	Open-NL	0.0	7.2	20.4	23.0	24.3	12.8
A4	NAL-	0.0	0.0	0.9	12.5	21.8	21.8
	NL2						
	Open-NL	0.0	0.0	6.6	18.4	32.7	18.9
A5	NAL-	0.0	0.0	6.6	20.7	27.1	26.6
	NL2						
	Open-NL	0.0	0.0	11.2	27.6	38.8	23.5
A6	NAL-	22.5	24.2	32.9	35.6	41.4	42.6
	NL2						
	Open-NL	23.6	31.4	37.8	39.1	43.2	34.0
A7	NAL-	34.7	34.6	34.7	34.8	35.0	35.1
	NL2						
	Open-NL	37.5	37.5	37.5	37.5	37.5	37.5

474 To establish a standardized comparative baseline, Open-NL’s algorithmic sensitivity is
 475 evaluated across seven canonical audiometric profiles (A1–A7). Profiles A1–A5 represent the
 476 standard sensorineural configurations utilized by Johnson & Dillon (2011) (derived from the
 477 foundational profiles of Byrne), spanning mild-sloping (A1), reverse-slope (A2), and severe
 478 to profound (A4, A5) pathologies. Profiles A6 and A7 expand this set to demonstrate the

framework’s mechanical handling of mixed and pure-conductive pathologies. While evaluating a large-scale real-world corpus (e.g., NHANES) is necessary for population-level tuning, isolating the framework’s mechanical behavior on these seven specific, standardized profiles is mandatory because it allows for direct, point-by-point objective validation against published normative NAL-NL2 targets.

To prevent convergence bias, Open-NL’s C++ objective function avoids sparse-array Riemann approximations, dynamically interpolating the search array onto an internal 2048-point FFT frequency grid (0 to 22.05 kHz). For mild-to-moderate losses (A1, A3), Open-NL expands soft speech (50 dB inputs) slightly more aggressively than NAL-NL2 to maximize audibility within the safe physiological envelope, while compressing higher-level inputs to maintain loudness parity (Figure 3, Figure 4).

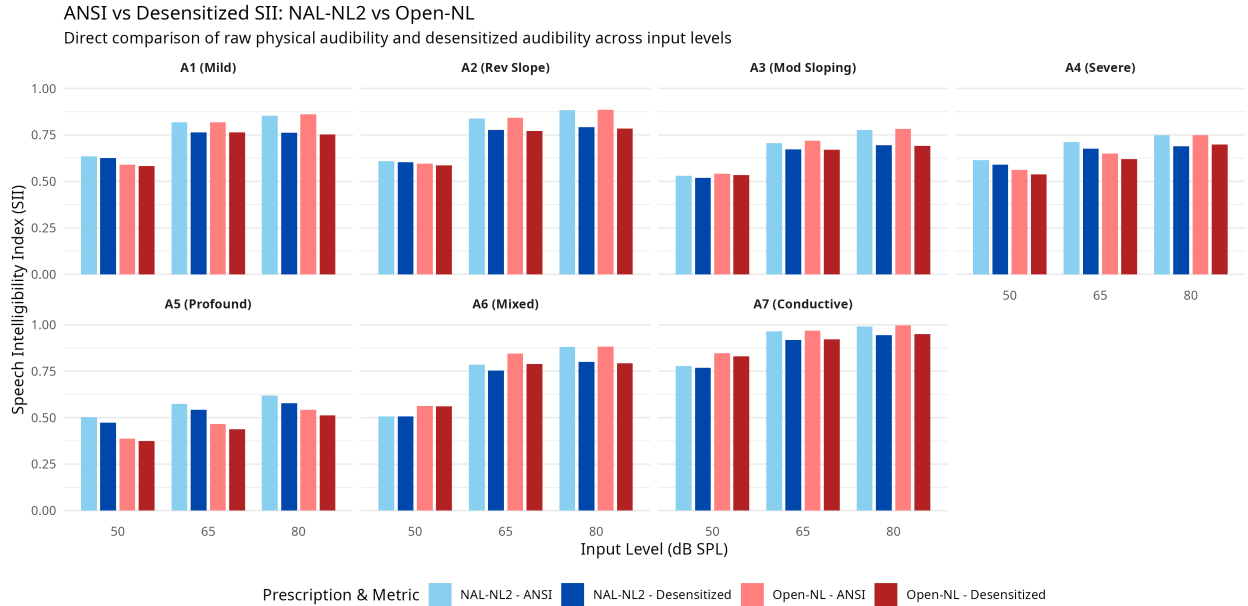


Figure 3. Comparison of ANSI SII (Raw Physical Audibility) vs Desensitized SII for NAL-NL2 and Open-NL across 50, 65, and 80 dB SPL Inputs.

Prescriptive Gain Targets: Open-NL vs NAL-NL2

Inputs at 50, 65, and 80 dB SPL across seven canonical audiometric profiles

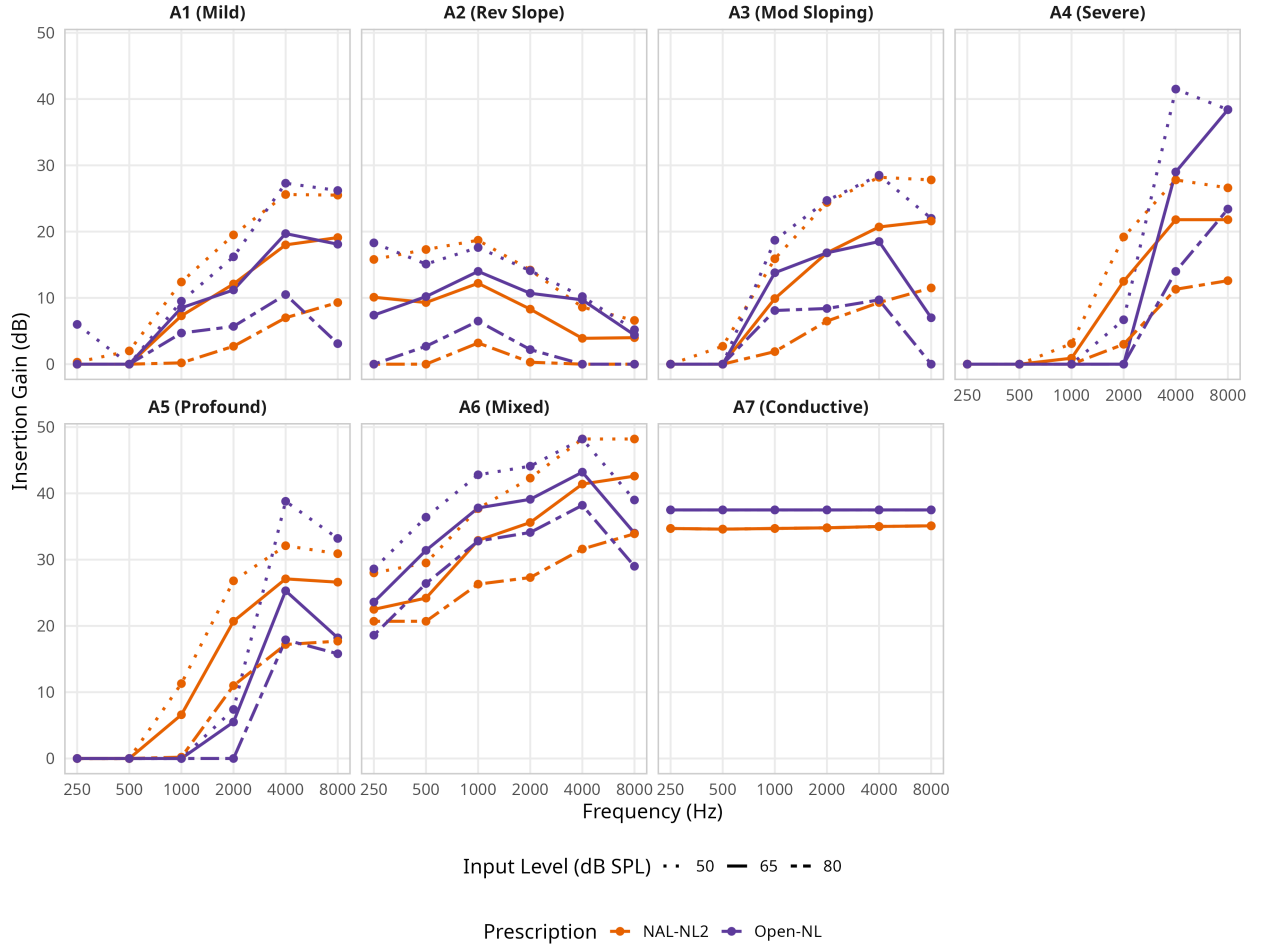


Figure 4. Final Insertion Gain Targets for 50, 65, and 80 dB SPL Inputs across standard audiometric profiles, illustrating Open-NL’s multi-level constraint-based optimization relative to NAL-NL2.

For severe and profound losses (A4, A5), unmodified SII maximization drives substantial high-frequency gain. Although Open-NL integrates desensitization penalties to temper this drive, it still prescribes substantially more high-frequency gain than NAL-NL2 (e.g., +10.9 dB at 4 kHz for A4, and +11.7 dB for A5 at 65 dB SPL inputs; Table IV). In listeners with severe loss, reduced spectral resolution, elevated hearing thresholds, and cochlear dead regions

502 account for comparable shares of speech recognition variance, with dead regions specifically
 503 blunting the benefit of restored high-frequency audibility (Ching, Dillon, & Byrne, 1998;
 504 Baer, Moore, & Kluk, 2002; Vestergaard, 2003; Souza et al., 2018; Moualed, Humphries, &
 505 Ramsden, 2018). While high prescribed gain increases physical audibility on paper, it severely
 506 degrades perceptual clarity if suprathreshold distortion is unmodeled (Margolis et al., 2025).
 507 However, enforcing blanket high-frequency suppression based purely on pure-tone audiograms
 508 would penalize the majority of candidates who benefit from audibility (Cox et al., 2011, 2012;
 509 Pepler et al., 2015). Furthermore, as Engler, Digeser, and Hoppe (2026) demonstrated, aided
 510 speech recognition remains practically insufficient in ears above ~ 80 dB HL regardless of
 511 prescribed gain. This tension underscores why high-frequency boundaries must be tied to
 512 confirmed dead-region diagnostics (e.g., TEN tests) and individualized distortion limits rather
 513 than static audiograms.

514 For conductive and mixed losses (A6, A7), Open-NL separates the mechanical attenuation
 515 of the middle ear from sensorineural cochlear distortion, restricting desensitization penalties
 516 strictly to sensorineural thresholds. In profile A7 (pure conductive loss with a 50 dB air-bone
 517 gap), the output is fully deterministic: the 75% ABG restoration rule mandates an exact, flat
 518 37.5 dB of linear gain across frequencies and levels. Because the optimizer contributes nothing
 519 to this solution and both formulas mechanically converge on ANSI SII 0.97, A7 is included
 520 in the tables strictly as an arithmetic sanity check rather than a comparative optimization
 521 finding. Crucially, this 75% restoration rule is an engineering choice adapted from clinical
 522 conventions (Johnson, 2013a; Scollie et al., 2005) to prevent hardware saturation, rather
 523 than an empirical preference optimum. This contrasts with well-supported heuristic targets
 524 like the 3.0:1 Compression Ratio bound (Stage 12), which is directly grounded in extensive

empirical psychoacoustic data (Souza, 2002; Souza et al., 2006). However, to enforce this bound safely during unconstrained optimization, Open-NL applies the 3.0:1 constraint both as a soft objective penalty (P_{cr}) to guide the optimizer, and as a strict post-optimization hard clamp. This dual constraint structure ensures that the raw drive to maximize SII in profound profiles never violates empirical psychoacoustic limits. For reverse-slope losses (A2), the SD-LFP constraint successfully limits low-frequency over-amplification, demonstrating how integrated constraints stabilize complex objective landscapes.

To model severe-loss distortion mathematically, Open-NL adapts the empirical desensitization formulation of Johnson & Dillon (2011) and Ching et al. (1998). Crucially, the engine isolates the pure sensorineural component ($T_{hl} = \max(0, T'_i - J_i)$) by subtracting the air-bone gap (J_i), and corrects a historical flaw in ANSI S3.5 implementations by restricting the internal cochlear noise floor calculation strictly to sensorineural loss ($X'_i = X_i + \max(0, T'_i - J_i)$), preventing conductive attenuation from falsely inflating internal noise. While the rigid clinical formula ($K'_{complete} = (K_i^p + m^p)^{1/p}$) introduces non-differentiable step boundaries that stall simplex optimizers, Open-NL’s optimizer evaluates intermediate solutions against a continuous mathematical relaxation:

$$K'_{smoothed} = K_i \cdot m, \quad \text{where } m = \frac{1}{1 + e^{0.075(T_{hl} - 66)}} \quad (1)$$

where K_i is raw audibility and m is the maximum asymptotic audibility limit directly extracted from Ching et al. (1998). This continuous relaxation permits smooth gradient descent. While the maximum discrepancy between the relaxation and the full piecewise function ($\max |K'_{smoothed} - K'_{complete}|$) reaches up to 0.20 raw band audibility units at intermediate

thresholds ($T_{hl} \approx 67$ dB HL), finalized targets are rigorously post-scored against the rigid piecewise $K'_{complete}$ formulation to ensure objective integrity (detailed fully in Stage 7 and Stage 12 of the Supplementary Material).

C. Clinical Validation Protocols and Falsifiable Predictions

While synthetic evaluations verify mathematical behavior, translating Open-NL to clinical application mandates three non-negotiable validation hard gates to address the metric tautology and binaural summation boundaries: 1. **Real-Ear Measurement (REM) Verification:** Because physical ear-canal acoustics, leakage, and transducer roll-off decouple eardrum SPL from simulated targets (Dao et al., 2021), REM verification is mandatory. Empirical evidence robustly demonstrates that REM-verified fittings significantly outperform unverified first-fits on speech recognition and patient preference (Valente et al., 2018; Almuftarij, Dillon, & Munro, 2021). 2. **Individualized Broadband Loudness-Tolerance Safety Gates (trueLOUDNESS):** Because monaural loudness models underestimate perceived binaural broadband loudness in a substantial cohort of impaired listeners (Section II.A), clinical translation cannot rely on audiogram-derived monaural ceilings. Prior to any behavioral testing, individualized binaural broadband loudness scaling (e.g., trueLOUDNESS procedures; Oetting et al., 2016, 2017) or rigorous Uncomfortable Loudness Level (UCL) verification must be administered to establish subject-specific safety caps and prevent acoustic trauma. 3. **Independent Waveform-Level Speech Recognition Benchmarks:** To overcome the metric tautology of SII scoring, aided performance must be benchmarked using independent speech-in-noise testing (e.g., matrix sentence tests or WIN/HINT) alongside computational HASPI/HASQI modeling (Kates & Arehart, 2022). These evaluations must

reference conservative clinical controls (such as DSL v5.0 or NAL-NL2), which empirical literature robustly favors in the 50–80 dB HL range (Mueller, 2005; Engler, Digeser, & Hoppe, 2026).

Beyond safety protocols, this framework yields a concrete, falsifiable clinical prediction: because Open-NL’s uncalibrated A4 and A5 high-frequency targets exceed NAL-NL2 by roughly 11 dB at 4000 Hz, they push far beyond historical comfort boundaries (Keidser et al., 2012a; Denk et al., 2025). We offer the following operational hypothesis: If adult listeners with A4 or A5 audiometric profiles are fitted with real-ear verified Open-NL targets, >80% will exhibit immediate categorical loudness rejection—operationally defined as a rating of 6 (“Loud”) or 7 (“Uncomfortably Loud”) on the 7-point Categorical Loudness Scaling (CLS) procedure (ISO 16832)—when presented with continuous broadband speech (e.g., ISTS) at 65 and 80 dB SPL, relative to a matched NAL-NL2 baseline. Empirically quantifying this rejection threshold will provide the ground-truth data required to constrain distortion-aware objective functions in future stochastic calibrations.

IV. CONCLUSION

Open-NL provides a transparent, modular computational testbed for modeling, ablating, and evaluating WDRC prescriptive heuristics natively within R. By coupling an explicitly defined mathematical pipeline with an embedded C++ specific-loudness engine, the package enables researchers to systematically inspect the trade-offs between audibility and physiological loudness without relying on closed-source clinical software. As the framework evolves, it provides the computational substrate needed to evaluate emerging multi-profile rationales

588 such as NAL-NL3 (Kitterick, Zakis, & Edwards, 2026a) and to integrate individualized
589 broadband loudness summation metrics (Denk et al., 2025).

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593 toolkit.

594 **AUTHOR DECLARATIONS**

595 **Conflict of Interest**

596 The author declares no conflicts of interest.

597 **Ethics Approval**

598 The author declares that no animal subjects or human participants were involved in the
599 development, theoretical simulation, or mathematical validation presented in this research.

600 **DATA AVAILABILITY**

601 The source code for the SII package, the Open-NL prescriptive algorithm, complete
602 parameter specifications, and all associated datasets and benchmarking scripts are openly
603 available in the public repository at <https://github.com/r-gregmisc/SII> (v1.2.4; Git commit
604 b2b5ce0; Archival DOI: 10.5281/zenodo.14963842; License: GPL-3.0). Standalone replication
605 scripts generating all figures, tables, and sensitivity sweeps reported in this manuscript are

located in the `reproducibility_scripts/` directory.

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Supplementary Material: Open-NL Algorithmic Pipeline and Complete Parameter Specification

This supplementary document provides the exact mathematical formulations, closed-form piecewise functions, and complete numerical parameter specifications governing the twelve internal processing stages of the Open-NL algorithmic cascade.

S.I. The Twelve-Stage Execution Cascade

Open-NL operates as a multi-stage parameterized shape generator. Rather than relying on static compiled lookup tables, Open-NL calculates target insertion gains dynamically through a series of twelve explicitly defined, cascaded mathematical modules. Each step in the gain derivation process is exposed natively in R, available for researchers to inspect, modify, and tune.

Terminology Note: Throughout this framework, the algorithm utilizes two distinct discomfort predictors for different theoretical purposes: an HL-domain “LDL” (Loudness Discomfort Level) predictor used for estimating clinical audiometric dynamic range (Stage 8), and an SPL-domain “UCL” (Uncomfortable Loudness Level) predictor for establishing physical device saturation limits (Stage 11). A visual flowchart mapping each predictor to its downstream algorithm function is provided in Diagram 1.

DIAGRAM 1. Open-NL Discomfort Predictors

[CLINICAL DYNAMIC RANGE]	[PHYSICAL DEVICE LIMITS]
v	v
LDL Predictor	UCL Predictor
(dB HL)	(dB SPL)
$100 + \max(0, \text{HTL} - 40) * 0.5 + \text{Loss_cond}$	$105 + 0.5 * \max(0, \text{HTL} - 20)$
v	v
Dynamic Range "Squeeze"	Maximum Power Output (MPO)
(Insertion Gain Attenuation)	(Saturation Limit & CR Calc)

=====

The twelve modules operate in a strictly defined cascaded execution order to prevent unintended interactions between additive boosters and soft limiters:

1. **Stage 1: Conductive Component Separation & Dynamic Range Baseline**
(Section II.D)
2. **Stage 2: Decoupled Half-Gain Base Anchor Calculation** (Section II.E)
3. **Stage 3: Experience-Level Shaping & Log-Frequency C_{vals} Interpolation**
(Section II.F)
4. **Stage 4: Reverse-Slope Low-Frequency Attenuation Floor** (Section II.G)

5. **Stage 5: Slope-Dependent Low-Frequency Penalty (SD-LFP) with Profound HF Bypass** (Section II.H)
6. **Stage 6: Severe-Loss Audibility Booster** (Section II.I)
7. **Stage 7: Soft-Compression High-Frequency Desensitization** (Section II.J)
8. **Stage 8: Dynamic Range Mapping (LDL Squeeze)** (Section II.K)
9. **Stage 9: Transducer Bandwidth Roll-off** (Section II.L)
10. **Stage 10: Multi-Channel WDRC Mapping, Input/Output Pivot, and Demographic Adjustments** (Section II.M)
11. **Stage 11: Acoustic Venting, Coupling Loss, and Receiver Saturation Limits** (Section II.N)
12. **Stage 12: Embedded Nelder-Mead Simplex Optimization & Physiological Loudness Ceilings** (Section II.O)

S.I.1. Stage 1: Conductive Component Separation & Dynamic Range Baseline

The algorithm first decomposes total hearing threshold levels (HTL) into their physiological constituents: the sensorineural threshold (HTL_{sn}) and the conductive component ($Loss_{cond}$, corresponding to the air-bone gap, ABG):

$$Loss_{cond}(f) = ABG(f) \quad (2)$$

$$\text{HTL}_{sn}(f) = \max(0, \text{HTL}(f) - \text{Loss}_{cond}(f)) \quad (3)$$

To ensure that equal-loudness reshaping penalties apply only to sensorineural pathology, the algorithm computes a frequency-specific sensorineural ratio $R_{sn}(f)$:

$$R_{sn}(f) = \begin{cases} \frac{\text{HTL}_{sn}(f)}{\text{HTL}(f)}, & \text{if } \text{HTL}(f) > 0 \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

Pure conductive losses ($\text{Loss}_{cond} = \text{HTL}$) have normal outer and inner hair cell functioning and intact basilar membrane mechanics; thus, $R_{sn} = 0$, bypassing cochlear recruitment and equal-loudness correction arrays.

S.I.2. Stage 2: Decoupled Half-Gain Base Anchor Calculation

The foundational WDRC anchor for average conversational speech (65 dB SPL) is derived using a frequency-specific adaptation of the half-gain rule (Lybarger, 1944). Unlike linear prescriptions that anchor to a broadband Pure Tone Average (PTA), this anchor is decoupled frequency-by-frequency:

$$G_{base}(f) = \alpha \cdot \text{HTL}_{sn}(f) + C_{interp}(f) \quad (5)$$

where $\alpha = 0.46$ is the nominal half-gain multiplier (loosely derived from Byrne & Dillon, 1986), and $C_{interp}(f)$ is the frequency-shaping correction array derived in Stage 3.

772 **S.I.3. Stage 3: Experience-Level Shaping & Log-Frequency C_{vals}**

773 **Interpolation**

774 To account for listener acclimatization and preferred listening levels (Keidser et al.,
 775 2012a), Open-NL modulates the frequency-shaping array C_{interp} across eight discrete anchor
 776 frequencies:

$$\mathbf{f}_c = [250, 500, 1000, 2000, 3000, 4000, 6000, 8000] \text{ Hz} \quad (6)$$

777 The discrete shaping vectors $\mathbf{C}$ are defined as: - **Experienced Users** (standard baseline):

$$\mathbf{C}_{exp} = [-8, -1, 3, 1, 0, 0, 0, 0] \text{ dB} \quad (7)$$

778 - **New Users** (nominal -3 dB reduction to combat occlusion and sharpness):

$$\mathbf{C}_{new} = \mathbf{C}_{exp} - 3 = [-11, -4, 0, -2, -3, -3, -3, -3] \text{ dB} \quad (8)$$

779 - **Power Users** (prioritizes raw audibility over acoustic comfort):

$$\mathbf{C}_{power} = \mathbf{C}_{exp} + 3 = [-5, +2, 6, 4, 3, 3, 3, 3] \text{ dB} \quad (9)$$

780 For any calculation frequency f , $C_{interp}(f)$ is evaluated via piecewise log-linear interpola-
 781 tion across $\mathbf{f}_c$ and scaled by the sensorineural proportion $R_{sn}(f)$:

$$C_{interp}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_c, \mathbf{C}) \cdot R_{sn}(f) \quad (10)$$

782

783 **S.I.4. Stage 4: Reverse-Slope Low-Frequency Attenuation Floor**

784 For reverse-slope configurations (where low frequencies are significantly worse than high
 785 frequencies), attempting to fully restore low-frequency audibility risks upward spread of
 786 masking, where high-energy low-frequency vowels mask low-energy high-frequency consonants.
 787 The algorithm calculates the low-to-high sensorineural threshold difference:

$$\text{Diff}_{rs} = \max\left(0, \overline{\text{HTL}}_{sn, \leq 500} - \overline{\text{HTL}}_{sn, \geq 2000}\right) \quad (11)$$

788 where $\overline{\text{HTL}}_{sn, \leq 500}$ is the mean threshold across $f \leq 500$ Hz, and $\overline{\text{HTL}}_{sn, \geq 2000}$ is the mean
 789 threshold across $f \geq 2000$ Hz. If $\text{Diff}_{rs} > 15$ dB, a transition factor $RS_{factor} \in [0, 1]$ is
 790 computed:

$$RS_{factor} = \max\left(0, \min\left(1, \frac{\text{Diff}_{rs} - 15}{20}\right)\right) \quad (12)$$

791 A log-linear low-frequency taper $W_{LF}(f)$ is applied below 1000 Hz:

$$W_{LF}(f) = \max\left(0, \min\left(1, 1 - \frac{\log_{10}(f) - \log_{10}(250)}{\log_{10}(1000/250)}\right)\right) \quad (13)$$

792 Across octave bands, $W_{LF}(f)$ evaluates to 1.0 at 250 Hz, 0.5 at 500 Hz, and 0.0 at
 793 $f \geq 1000$ Hz. The frequency-shaping array is then dynamically flattened toward a -10 dB

794 floor:

$$C'_{interp}(f) = C_{interp}(f) \cdot (1 - RS_{factor} \cdot W_{LF}(f)) + (-10 \cdot RS_{factor} \cdot W_{LF}(f)) \quad (14)$$

795

796 **S.I.5. Stage 5: Slope-Dependent Low-Frequency Penalty (SD-LFP)**

797 **with Profound HF Bypass**

798 For sloping high-frequency losses, applying excessive low-frequency gain causes normal
 799 low frequencies to dominate broadband loudness. Open-NL computes the high-frequency
 800 slope:

$$\text{Slope} = \max\left(0, \overline{\text{HTL}}_{sn, \geq 2000} - \overline{\text{HTL}}_{sn, \leq 500}\right) \quad (15)$$

801 To operationalize the empirical findings of Byrne, Parkinson, and Newall (1990)—who
 802 showed that listeners with high-frequency losses exceeding 70–95 dB HL require low-frequency
 803 speech cues—Open-NL incorporates a **Profound High-Frequency Bypass**:

$$PF_{bypass} = \max\left(0, \min\left(1, \frac{95 - \overline{\text{HTL}}_{sn, \geq 2000}}{25}\right)\right) \quad (16)$$

804 The low-frequency penalty is scaled over a 20 dB slope window, capped at 15 dB:

$$\text{LF}_{penalty} = \max\left(0, \min\left(1, \frac{\text{Slope} - 15}{20}\right)\right) \cdot 15 \cdot PF_{bypass} \quad (17)$$

805 The penalty is then tapered below 1000 Hz using $W_{LF}(f)$:

$$C''_{interp}(f) = C'_{interp}(f) - (LF_{penalty} \cdot W_{LF}(f)) \quad (18)$$

806 The resulting baseline target at 65 dB SPL is:

$$G_{65}^{(0)}(f) = 0.46 \cdot HTL_{sn}(f) + C''_{interp}(f) \quad (19)$$

807 Ablation of SD-LFP Constraints

808 Table S1 provides the reference audiometric profiles (A1–A7, Johnson & Dillon, 2011).

809 Table S2 demonstrates the mechanical impact of the SD-LFP constraint on the initial heuristic
810 seeds.

811 **TABLE S1. Reference Audiometric Profiles (Adapted from Johnson & Dillon,**
812 **2011).** *Thresholds are in dB HL. Profiles A1–A5 are purely sensorineural. A6 is mixed (30*
813 *dB ABG). A7 is conductive (50 dB ABG).*

Profile	Type	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz	ABG
A1	Mild	15	20	30	40	50	60	0 dB
A2	Reverse slope	60	50	40	30	20	15	0 dB
A3	Moderately slop- ing	10	20	40	50	55	60	0 dB
A4	Severe	0	0	10	40	70	80	0 dB

Profile	Type	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz	ABG
A5	Profound	10	10	20	60	80	100	0 dB
A6	Mixed	50	55	60	65	75	80	30 dB
A7	Conductive	50	50	50	50	50	50	50 dB

TABLE S2. Ablation of SD-LFP Constraints on Optimizer Seed (65 dB SPL Input).

	Unconstrained	Unconstrained	SD-LFP Seed	SD-LFP Seed
Profile	Seed SII	Seed Sones	SII	Sones
A1 (Flat mod)	0.86	7.2	0.86	7.2
A2 (Reverse)	0.88	6.3	0.88	6.0
A3 (Mod sloping)	0.79	6.8	0.79	6.8
A4 (Severe)	0.81	6.9	0.81	6.9
A5 (Profound)	0.68	7.1	0.68	6.0
A6 (Mixed)	0.82	3.0	0.82	3.0
A7 (Conductive)	0.97	1.0	0.97	1.0

S.I.6. Stage 6: Severe-Loss Audibility Booster

To overcome inner hair cell loss in severe impairments, an opt-in, bounded **Severe-Loss Booster** can be applied:

$$G_{slb}(f) = B_{en} \cdot 0.15 \cdot \max(0, \min(80, \text{HTL}_{sn}(f)) - T_{onset}) \quad (20)$$

820 where $B_{en} \in \{0, 1\}$ (default: 0, disabled), and $T_{onset} = 70$ dB HL (conservative default)
821 or 60 dB HL (aggressive ablation mode). The intermediate target is:

$$G_{65}^{(1)}(f) = G_{65}^{(0)}(f) + G_{slb}(f) \quad (21)$$

822

823 **S.I.7. Stage 7: Soft-Compression High-Frequency Desensitization**

824 To prevent unconstrained audibility maximization from prescribing intolerable high-
825 frequency gain in steeply sloping losses, Open-NL applies a dynamic soft-compression envelope
826 (L_{gain}):

$$L_{gain}(f) = 30 + 0.4 \cdot \max(0, \text{HTL}_{sn}(f) - 60) \text{ (Power Users: } L_{gain}(f) = 40 + 0.5 \cdot \max(0, \text{HTL}_{sn}(f) - 60)) \quad (22)$$

827 (*Note: For patients in “Moderate” or “High” distortion categories, L_{gain} is reduced by 10*
828 *dB*).

829 Excess gain above this dynamic limit is:

$$\text{Excess}(f) = \max(0, G_{65}^{(1)}(f) - L_{gain}(f)) \quad (23)$$

830 The sloping factor $S_{factor}(f)$ relative to the best low-frequency threshold ($\text{HTL}_{bestlow} =$
831 $\min_{f' \leq 1000} \text{HTL}_{sn}(f')$) and the high-frequency fade-in weight $W_{hf}(f)$ are:

$$S_{factor}(f) = \max \left(0, \min \left(1, \frac{\text{HTL}_{sn}(f) - \text{HTL}_{bestlow} - 25}{20} \right) \right) \quad (24)$$

$$W_{hf}(f) = \max \left(0, \min \left(1, \frac{f - 2000}{2000} \right) \right) \quad (25)$$

832 Applying 2:1 soft compression to the excess yields:

$$G_{65}^{(2)}(f) = G_{65}^{(1)}(f) - (W_{hf}(f) \cdot S_{factor}(f) \cdot 0.50 \cdot \text{Excess}(f)) \quad (26)$$

833 If explicit cochlear dead regions (f_{e_hf} , f_{e_lf}) or distortion categories (Margolis et al.,
834 2025) are defined, steep parametric roll-offs are applied:

$$G_{65}^{(2)}(f) \leftarrow G_{65}^{(2)}(f) - \max \left(0, \log_2 \left(\frac{f}{1.7 f_{e_hf}} \right) \right) \cdot 30 - \max \left(0, \log_2 \left(\frac{0.57 f_{e_lf}}{f} \right) \right) \cdot 30 \quad (27)$$

835

836 **S.I.8. Stage 8: Dynamic Range Mapping (LDL Squeeze)**

837 To accommodate reduced clinical dynamic ranges (DSL v5.0 philosophy; Scollie et al.,
838 2005), Open-NL predicts an HL-domain Loudness Discomfort Level:

$$\text{LDL}_{pred}(f) = 100 + 0.5 \cdot \max(0, \text{HTL}_{sn}(f) - 40) + \text{Loss}_{cond}(f) \quad (28)$$

839 When measured $\text{LDL}_{meas}(f)$ is lower than predicted, the dynamic range discrepancy
840 $\Delta_{LDL}(f)$ is quantified:

$$\Delta_{LDL}(f) = \text{LDL}_{meas}(f) - \text{LDL}_{pred}(f) \quad (29)$$

841 Target gain is attenuated by 0.2 dB per dB of dynamic range squeeze:

$$G_{65}^{(3)}(f) = G_{65}^{(2)}(f) + (0.2 \cdot \Delta_{LDL}(f)) \quad (30)$$

842 Simultaneously, the baseline compression ratio increases by +0.02 per dB of squeeze
843 (Stage 10).

844

845 **S.I.9. Stage 9: Transducer Bandwidth Roll-off**

846 Acoustic transducers physically struggle to reproduce frequencies at the extremes of the
847 spectrum (≤ 250 Hz and ≥ 6000 Hz), where massive gain leads to distortion and feedback.
848 Open-NL applies a continuous fractional bandwidth roll-off multiplier $M_{bw}(f)$ defined over
849 seven anchor frequencies:

$$\mathbf{f}_{bw} = [250, 500, 1000, 2000, 4000, 6000, 8000] \text{ Hz} \quad (31)$$

850

$$\mathbf{w}_{bw} = [0.7, 1.0, 1.0, 1.0, 1.0, 0.8, 0.5] \quad (32)$$

851 The closed-form continuous multiplier is obtained via piecewise log-linear interpolation:

$$M_{bw}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_{bw}, \mathbf{w}_{bw}) \quad (33)$$

$$G_{65}^{(4)}(f) = G_{65}^{(3)}(f) \cdot M_{bw}(f) \quad (34)$$

852

853 **S.I.10. Stage 10: Multi-Channel WDRC Mapping, Input/Output**
854 **Pivot, and Demographic Adjustments**

855 Target gain at arbitrary overall input level L_{in} (e.g., 50, 65, 80 dB SPL) is computed using
856 a multi-channel wide dynamic range compression architecture pivoted around the Long-Term
857 Average Speech Spectrum (LTASS).

- 858 1. **Speech Spectrum Pivot:** Let $P(f)$ be the critical-band normal speech level at 65
859 dB SPL overall. The band-specific input level is:

$$L_{band}(f) = P(f) + (L_{in} - 65) \quad (35)$$

- 860 2. **Dynamic Compression Ratio Calculation:**

$$\text{CR}_{base}(f) = 1 + \frac{\max(0, \text{HTL}_{sn}(f) - 20)}{40} - (0.02 \cdot \Delta_{LDL}(f)) \quad (36)$$

861 For severe loss ($\text{HTL}_{sn} > 65$ dB HL), compression reduces back toward linear to preserve

the temporal speech envelope (Keidser et al., 2007):

$$W_{freq}(f) = \max \left(0, \min \left(1, \frac{f - 500}{2500} \right) \right) \quad (37)$$

$$CR_{loud}(f) = CR_{base}(f) - \left(\frac{\max(0, HTL_{sn}(f) - 65)}{30} \cdot (1.5 - 0.5 \cdot W_{freq}(f)) \right) \quad (38)$$

$CR_{loud}(f)$ is clamped between 1.0 and a maximum $CR_{max}(f) = 1.5 + 0.9 \cdot W_{freq}(f)$ (further relaxed for age > 60). In Comfort in Noise (CIN) mode, CR_{loud} is clamped to ≤ 1.5 .

(Note on Evidentiary Asymmetry: The 1.5:1 CIN clamp is an asserted engineering heuristic inspired by NAL-NL3 to curb listening fatigue in noise, lacking direct empirical derivation. In contrast, the global upper ceiling on compression ratios ($\leq 3.0 : 1$) is directly supported by extensive psychoacoustic literature [Souza, 2002; Souza et al., 2006], which demonstrates that compression ratios exceeding 3.0:1 cause severe temporal envelope flattening and acoustic contrast degradation).

3. Variable Compression Threshold (CT): Overall CT ($CT_{overall}$) scales from 30 to 45 dB SPL as a function of threshold. The band-level threshold is $CT_{band}(f) = P(f) + (CT_{overall} - 65)$ (reduced by 10 dB in CIN mode).

4. Piecewise Linear-Compressive Spline: Gain at the compression threshold is:

$$G_{CT}(f) = \begin{cases} G_{65}^{(4)}(f), & \text{if } CT_{band}(f) > P(f) \\ G_{65}^{(4)}(f) + (P(f) - CT_{band}(f)) \left(1 - \frac{1}{CR_{loud}(f)} \right), & \text{if } CT_{band}(f) \leq P(f) \end{cases} \quad (39)$$

877 The level-dependent WDRC target gain is:

$$G_{target}(f, L_{in}) = \begin{cases} G_{CT}(f), & \text{if } L_{band}(f) \leq CT_{band}(f) \\ G_{CT}(f) - (L_{band}(f) - CT_{band}(f)) \left(1 - \frac{1}{CR_{loud}(f)}\right), & \text{if } L_{band}(f) > CT_{band}(f) \end{cases} \quad (40)$$

878 5. Demographic Adjustments:

$$G_{target}(f, L_{in}) \leftarrow G_{target}(f, L_{in}) + \Delta_{gender} + \Delta_{config} + \Delta_{exp} \quad (41)$$

879 where $\Delta_{gender} = -1.5$ dB (female), $\Delta_{config} = +3.0$ dB (unilateral), and $\Delta_{exp} =$
880 $-\min(6.0, 0.3 \cdot \max(0, PTA_{500,1k,2k} - 40))$ for new users.

881 **TABLE S3. Effective Compression Ratios (50 to 80 dB SPL Inputs) across**
882 **A1-A7 Audiograms.** *Note: Dashes (-) indicate frequency regions where prescribed gain*
883 *is exactly 0 dB for both 50 and 80 dB SPL inputs (linear amplification, CR = 1.0). Note*
884 *that these values represent the emergent multi-level input/output ratios measured dynamically*
885 *between 50 and 80 dB SPL inputs, rather than the prescribed static channel CRs calculated*
886 *internally in Stage 10. To strictly enforce the 3.0:1 maximum bound, Open-NL applies dual*
887 *constraints: a soft objective penalty (P_{cr}) during optimization, followed by a strict hard clamp,*
888 *ensuring that pure intelligibility maximization never violates empirical psychoacoustic limits.*

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A1	NAL-	1.01	1.07	1.69	2.27	2.63	2.17
	NL2						

Profile	Formula	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
A1	Open-NL	1.04	-	1.42	1.36	1.66	1.66
A2	NAL-	2.11	2.36	2.07	1.86	1.40	1.28
	NL2						
A2	Open-NL	1.50	1.78	2.10	1.73	1.18	-
A3	NAL-	-	1.10	1.88	2.48	2.70	2.19
	NL2						
A3	Open-NL	-	1.03	1.71	1.59	3.00	3.00
A4	NAL-	-	-	1.12	2.17	2.22	1.88
	NL2						
A4	Open-NL	-	-	-	1.45	2.51	1.50
A5	NAL-	-	-	1.59	2.11	1.99	1.79
	NL2						
A5	Open-NL	-	-	-	1.83	1.34	1.10
A6	NAL-	1.32	1.42	1.61	2.00	2.24	1.91
	NL2						
A6	Open-NL	1.16	1.27	1.38	1.47	1.64	1.71
A7	NAL-	1.00	1.00	1.00	1.00	1.00	1.00
	NL2						
A7	Open-NL	1.00	1.00	1.00	1.00	1.00	1.00

S.I.11. Stage 11: Acoustic Venting, Coupling Loss, and Receiver Saturation Limits

Real-Ear Aided Responses (REAR) are heavily influenced by acoustic coupling. Low-frequency leakage is modeled by log-frequency interpolation over anchor frequencies $\mathbf{f}_{vent} = [250, 500, 1000, 2000, 4000, 8000]$ Hz:

$$V_{loss}(f) = \text{interp}_{\log_{10}}(f, \mathbf{f}_{vent}, \mathbf{v}_c) \tag{42}$$

where $\mathbf{v}_c$ is the coupling-specific attenuation vector defined in Table S4.

TABLE S4. Acoustic Coupling Real-Ear Insertion Loss Vectors ($\mathbf{v}_c$, in dB).

Coupling Configu- ration	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
Custom	0	0	0	0	0	0
Oc- cluded						
Open	-35	-28	-15	-2	0	0
Dome						
Tulip	-25	-18	-5	0	0	0
Dome						
Double	-20	-10	0	0	0	0
Dome						

Coupling						
Configu-						
ration	250 Hz	500 Hz	1000 Hz	2000 Hz	4000 Hz	8000 Hz
Vent	-3	-1	0	0	0	0
(1mm						
Solid)						
Vent	-8	-2	0	0	0	0
(2mm						
Solid)						
Vent	-12	-4	0	0	0	0
(3mm						
Solid)						
Vent	-12	-3	0	0	0	0
(1mm						
Hollow)						
Vent	-22	-12	-5	-2	0	0
(2mm						
Hollow)						
Vent	-25	-15	-8	-4	0	0
(3mm						
Hollow)						

898 $\text{Loss}_{\text{cond}}(f)$ (Note: This 75% restoration fraction is a pragmatic engineering convention—
899 adapted from clinical practice to prevent excessive output demands and MPO clipping; Johnson,
900 2013a—without direct empirical derivation from listener preference). To prevent active anti-
901 phase cancellation demands and comb filtering, insertion gain is floored at $V_{\text{loss}}(f) - 10$
902 dB:

$$G_{\text{heuristic}}(f, L_{\text{in}}) = \max(G_{\text{target}}(f, L_{\text{in}}) + 0.75 \cdot \text{Loss}_{\text{cond}}(f) + V_{\text{loss}}(f), V_{\text{loss}}(f) - 10) \quad (43)$$

903 Hardware receiver limits (MPO/SSPL90) are established to avoid severe saturation
904 distortion:

$$\text{MPO}(f) = \min(120, 105 + 0.5 \cdot \max(0, \text{HTL}_{\text{sn}}(f) - 20) + \text{Loss}_{\text{cond}}(f)) \quad (44)$$

905

906 **S.I.12. Stage 12: Embedded Nelder-Mead Simplex Optimization &** 907 **Physiological Loudness Ceilings**

908 When `optimize = TRUE`, Open-NL adjusts the heuristic targets by minimizing an uncon-
909 strained multi-objective loss function via Nelder-Mead simplex search (`stats::optim`).

910 **Parameter Vector and Gain Formation**

911 The optimization parameter vector is $\boldsymbol{\delta} = [\delta_{250}, \delta_{500}, \delta_{1000}, \delta_{2000}, \delta_{4000}, \delta_{8000}]^T \in \mathbb{R}^6$. During
 912 evaluation, shifts are clamped:

$$\delta_{clamped,j} = \max(-60, \min(30, \delta_j)) \quad (45)$$

913 Candidate insertion gains $\mathbf{G} \in [0, 80]$ dB are interpolated to calculation frequencies:

$$G(f) = \max(0, \min(80, G_{heuristic}(f) + \delta_{clamped}(f))) \quad (46)$$

914 **Loss Function Formulation**

915 The Nelder-Mead solver minimizes:

$$\mathcal{L}(\boldsymbol{\delta}) = -100 \cdot \text{SII}_{desens}(\mathbf{G}) + P_{anchor} + P_{bounds} + P_{loud} + P_{spl} + P_{rough} + P_{order} + P_{cr} + P_{abg} \quad (47)$$

916 where SII_{desens} is the desensitized Speech Intelligibility Index calculated using the
 917 johnson2011_smoothed transfer function.

918 **Explicit Penalty Terms and Exact Weights (λ)**

919 1. **Anchor Penalty** ($\lambda_{anchor} = 0.1$):

$$P_{anchor} = 0.1 \sum_{j=1}^6 |\delta_j| \quad (48)$$

920 **2. Out-of-Bounds Penalty** ($\lambda_{bounds} = 1000.0$):

$$P_{bounds} = 1000.0 \left[\sum_{j=1}^6 \max(0, \delta_j - 30)^2 + \sum_{j=1}^6 \max(0, -\delta_j - 60)^2 \right] \quad (49)$$

921 **3. Physiological Loudness Ceiling Penalty** ($\lambda_{loud} = 2000.0$):

$$P_{loud} = 2000.0 \cdot \max(0, \text{Sones}_{MG04}(\mathbf{G}) - \text{Cap}) \quad (50)$$

922 *(Note: While gain-based penalties in this framework are squared to strongly penalize*
 923 *large deviations, the loudness penalty is explicitly linear. This is a deliberate choice:*
 924 *because sones inherently represent a compressive power-law transformation of physical*
 925 *acoustic energy, a linear penalty in the sone domain naturally exerts an exponentially*
 926 *growing restriction on the underlying insertion gain. Squaring the sone error introduces*
 927 *severe mathematical stiffness and destabilizes the simplex gradient.)* where Sones_{MG04}
 928 is the Moore & Glasberg (2004) specific-loudness integration computed via native
 929 C++. The dynamic U-shaped cap is interpolated across Pure Tone Average knots
 930 $\mathbf{PTA}_{knots} = [10, 32.5, 52.5, 72.5, 90]$ dB HL from level-specific sone vectors:

- 931 • $L_{in} = 50$ dB SPL: $\mathbf{K}_{50} = [1.5, 1.0, 0.8, 1.2, 1.2]$ sones
- 932 • $L_{in} = 65$ dB SPL: $\mathbf{K}_{65} = [7.0, 4.5, 4.0, 6.5, 6.0]$ sones
- 933 • $L_{in} = 80$ dB SPL: $\mathbf{K}_{80} = [20.0, 12.0, 10.0, 15.0, 14.0]$ sones

$$\text{Cap}_{base} = \text{interp}(\text{PTA}_{sn}, \mathbf{PTA}_{knots}, \mathbf{K}_{L_{in}}) \quad (51)$$

where PTA_{sn} is defined as the four-frequency pure-tone average of the sensorineural component at 500, 1000, 2000, and 4000 Hz. *Profile Adjustments:*

- Reverse-slope restriction ($L_{in} \geq 75$ dB SPL and low-to-high threshold difference > 10 dB): $\text{Cap} = \text{Cap}_{base} - 0.10 \cdot (\overline{\text{HTL}}_{\leq 500} - \overline{\text{HTL}}_{\geq 4000})$.
- Conductive air-bone gap adjustment: $\text{Cap} \leftarrow \text{Cap} - 0.25 \cdot \text{PTA}_{ABG}$ (if $L_{in} \geq 75$ dB SPL) or $+0.10 \cdot \text{PTA}_{ABG}$ (if $L_{in} < 75$ dB SPL).

4. Broadband SPL Saturation Ceiling Penalty ($\lambda_{spl} = 2000.0$):

$$P_{spl} = 2000.0 \cdot \max(0, \text{SPL}_{aided} - 110.0) \quad (52)$$

5. Spectral Roughness Penalty ($\lambda_{rough} = 0.5$):

$$P_{rough} = 0.5 \sum_{j=1}^5 (\delta_{j+1} - \delta_j)^2 \quad (53)$$

6. Inter-Level Order Monotonicity Penalty ($\lambda_{order} = 2000.0$): Enforces $G_{50} \geq G_{65}$

and $G_{80} \leq G_{65}$:

$$P_{order} = \begin{cases} 2000.0 \sum_{j=1}^6 \max(0, G_{65,j} - G_{50,j})^2, & \text{for } L_{in} = 50 \\ 2000.0 \sum_{j=1}^6 \max(0, G_{80,j} - G_{65,j})^2, & \text{for } L_{in} = 80 \end{cases} \quad (54)$$

7. Compression Ratio Ceiling Penalty ($\lambda_{cr} = 200.0$):

$$P_{cr} = \begin{cases} 200.0 \sum_{j=1}^6 \max(0, (G_{50,j} - G_{65,j}) - \Delta_{max,j})^2, & \text{for } L_{in} = 50 \\ 200.0 \sum_{j=1}^6 \max(0, (G_{65,j} - G_{80,j}) - \Delta_{max,j})^2, & \text{for } L_{in} = 80 \end{cases} \quad (55)$$

where $\Delta_{max,j} = 10.0 \cdot (1 - \text{ABG}_j / \max(0.001, \text{HTL}_j))$ for soft speech, bounding emergent compression ratios safely below 3.0:1 for sensorineural loss while enforcing linear amplification for conductive components. This 3.0:1 ceiling represents the best-supported parameter in the algorithm, firmly grounded in empirical psychoacoustic literature (Souza, 2002; Souza et al., 2006) demonstrating severe envelope flattening, loss of acoustic contrast, and speech-in-noise deficits for CRs $> 3.0 : 1$.

8. Air-Bone Gap Excursion Penalty ($\lambda_{abg} = 1.0$):

$$P_{abg} = \begin{cases} 1.0 \sum_{j=1}^6 \max(0, \delta_j)^2, & \text{if } \exists j, \text{Loss}_j > 0 \\ 0, & \text{otherwise} \end{cases} \quad (56)$$

Solver Controls and Convergence Tolerances

- **Algorithm:** Nelder-Mead Simplex via R's `stats::optim()`.
- **Iteration Ceiling:** `control = list(maxit = 800)`.
- **Convergence Tolerance:** Relative convergence tolerance `reltol = sqrt(.Machine$double.eps)` $\approx 1.49 \times 10^{-8}$.
- **Initial Simplex Seeding:** Seeding incorporates an audibility projection $\delta_{start} = \min(20, \max(0, \text{target_aided} - G_{heuristic}))$, where `target_aided` =

$\min(\text{UCL} - 5, \max(L_{in}, \text{HTL} + 10))$. Soft inputs receive +3 dB shift; loud inputs receive $\max(-10, \delta_{start} - 5)$ dB shift.

- **Multi-Start Strategy:** A 5-iteration multi-start routine (seeding the initial simplex with the NAL-R target and executing four additional randomized restarts with uniform random jitter $\delta_{start} \leftarrow \delta_{start} + \mathcal{U}(-5, +5)$) is deployed to stabilize numerical convergence, though Nelder-Mead inherently lacks formal global convergence guarantees.

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